COMSATS University Islamabad, Attock Campus

Department Of Computer Science

Course	*Information Security (Lab)*
Instructor	*Ms. Ambreen Gul*
Assignment No.	*01*

Student Details

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Explanation:

In this code, we shifted each letter forward for encryption and backward for decryption.
Spaces and special characters will remain the same.

Encryption Function:

- The encryption function takes two inputs: the original message , the shift value
- It creates an empty variable to store the encrypted result.
- It then checks each character of the message one by one, If the character is a lowercase letter, It finds its position in the lowercase alphabet, adds the shift value to that position, uses modulus 26 to make sure it stays within alphabet range and then adds the new shifted letter to the result.
- Same process is done using uppercase alphabet.
- If the character is a space or symbol, It keeps it unchanged.
- Finally, it returns the encrypted message.

Decryption Function:

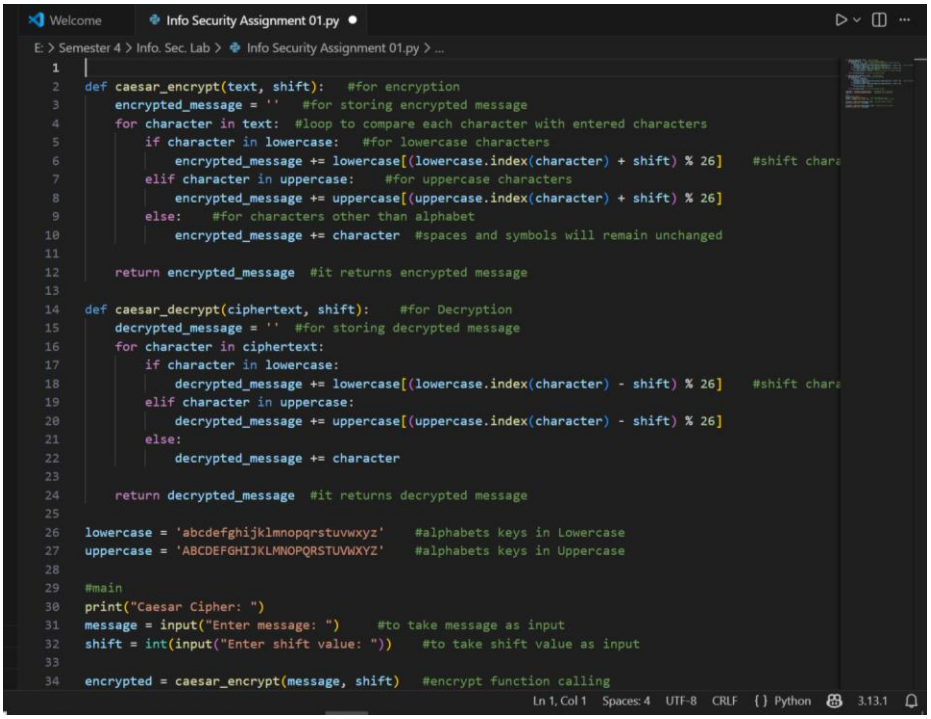
- The decryption function takes two inputs: the encrypted message, the same shift value.
- It creates an empty variable to store the decrypted result.
- It checks each character one by one.
- If lowercase, it finds position, subtracts shift, uses modulo 26, and adds original letter to result.
- If uppercase, same process using uppercase alphabet.
- If space or symbol, it stays unchanged.
- Finally, it returns the decrypted message.

Main:

- Prints the program title.
- Asks user to enter a message.

- Asks user to enter a shift value.
- Calls encryption function and prints encrypted message.
- Calls decryption function and prints original message.

Code Screenshot:



```python

def caesar_encrypt(text, shift):   #for encryption
    encrypted_message = ''   #for storing encrypted message
    for character in text:  #loop to compare each character with entered characters
        if character in lowercase:   #for lowercase characters
            encrypted_message += lowercase[(lowercase.index(character) + shift) % 26]    #shift chara
        elif character in uppercase:    #for uppercase characters
            encrypted_message += uppercase[(uppercase.index(character) + shift) % 26]
        else:    #for characters other than alphabet
            encrypted_message += character  #spaces and symbols will remain unchanged

    return encrypted_message  #it returns encrypted message

def caesar_decrypt(ciphertext, shift):    #for Decryption
    decrypted_message = ''  #for storing decrypted message
    for character in ciphertext:
        if character in lowercase:
            decrypted_message += lowercase[(lowercase.index(character) - shift) % 26]    #shift chara
        elif character in uppercase:
            decrypted_message += uppercase[(uppercase.index(character) - shift) % 26]
        else:
            decrypted_message += character

    return decrypted_message  #it returns decrypted message

lowercase = 'abcdefghijklmnopqrstuvwxyz'    #alphabets keys in Lowercase
uppercase = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ'    #alphabets keys in Uppercase

#main
print("Caesar Cipher: ")
message = input("Enter message: ")     #to take message as input
shift = int(input("Enter shift value: "))    #to take shift value as input

encrypted = caesar_encrypt(message, shift)   #encrypt function calling
```

```python

def caesar_decrypt(ciphertext, shift):    #for Decryption
    decrypted_message = ''  #for storing decrypted message
    for character in ciphertext:
        if character in lowercase:
            decrypted_message += lowercase[(lowercase.index(character) - shift) % 26]    #shift chara
        elif character in uppercase:
            decrypted_message += uppercase[(uppercase.index(character) - shift) % 26]
        else:
            decrypted_message += character

    return decrypted_message  #it returns decrypted message

lowercase = 'abcdefghijklmnopqrstuvwxyz'    #alphabets keys in Lowercase
uppercase = 'ABCDEFGHIJKLMNOPQRSTUVWXYZ'    #alphabets keys in Uppercase

#main
print("Caesar Cipher: ")
message = input("Enter message: ")     #to take message as input
shift = int(input("Enter shift value: "))    #to take shift value as input

encrypted = caesar_encrypt(message, shift)   #encrypt function calling
print("Encrypted Text:", encrypted)

decrypted = caesar_decrypt(encrypted, shift)   #decrypt function calling
print("Decrypted Text:", decrypted)

```

Output Screenshot:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\Semester 4\Info. Sec. Lab>  & 'c:\Users\LENOVO\AppData\Lo
.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle
Sec. Lab\Info Security Assignment 01.py'
Caesar Cipher:
Enter message: Information Security
Enter shift value: 3
Encrypted Text: Lqirupdwlrq Vhfxulwb
Decrypted Text: Information Security
PS E:\Semester 4\Info. Sec. Lab>
```